

1. The average teacher's salary in Australia is \$84,335. Suppose that the distribution is normal with standard deviation \$11,250. If we sample 100 teachers' salaries, what is the probability that the sample mean is less than \$80,000 per year? Write a decimal number with three decimal places.

**Solution**

```
pnorm(80000,84335,11250/sqrt(100))
#5.826029e-05
```

2. Consider a continuous random variable  $Y$  with pdf

$$f_Y(y) = \frac{2y}{k^2} \quad 0 \leq y \leq k.$$

For which value of  $k$  we have that  $\text{Var}(X) = 2$ ?

**Solution:** We have that

$$E(Y) = \int_0^k y \frac{2y}{k^2} dy = \frac{2k}{3}$$

$$E(Y^2) = \int_0^k y^2 \frac{2y}{k^2} dy = \frac{k^2}{2}$$

Then the variance is

$$\text{Var}(Y) = \frac{k^2}{2} - \frac{4k^2}{9} = \frac{k^2}{18}.$$

Having  $\text{Var}(Y) = 2$  implies that

$$\frac{k^2}{18} = 2 \rightarrow k = 6.$$

3. Suppose that  $X$  and  $Y$  have a bivariate uniform density over  $[0, 7]^2$

$$f_{XY}(x, y) = \begin{cases} c & 0 < x < 7, \quad 0 < y < 7 \\ 0 & \text{otherwise} \end{cases}$$

Find  $P(0 < X < 2, 0 \leq Y \leq 4)$ . Write a decimal number with three decimal places.

**Solution:**

$$1 = \int_0^7 \int_0^7 c dx dy = \int_0^7 cx|_0^7 dy = 7c \int_0^7 dy = 49c$$

therefore  $c = \frac{1}{49}$ .

Then

$$P(0 < X < 2, 0 \leq Y \leq 4) = \int_0^4 \int_0^2 \frac{1}{49} dx dy = \frac{8}{49} = 0.1632653$$

4. Suppose that  $f_{XY}(x, y) = \lambda^2 e^{-\lambda(x+y)}$ ,  $0 \leq x, 0 \leq y$ . Find  $E(X + Y)$ .

**Solution:** Since

$$f_{XY}(x, y) = \lambda^2 e^{-\lambda(x+y)} = (\lambda e^{-\lambda x}) \cdot (\lambda e^{-\lambda y})$$

we have that

$$E(X + Y) = E(X) + E(Y) = \frac{1}{\lambda} + \frac{1}{\lambda} = \frac{2}{\lambda}$$

5. A random sample of size 4 is drawn from a population that has a uniform distribution over the interval  $(0, 5)$ . The resulting order statistics are  $Y_{(1)}, Y_{(2)}, Y_{(3)}, Y_{(4)}$ . Compute the probability  $P(Y_{(3)} > 2)$ . Write a decimal number with three decimal places.

**Solution:** The cdf of a uniform random variable is

$$F_Y(y) = \frac{y - a}{b - a} = \frac{y}{5}.$$

The density of  $Y_{(3)}$  is

$$f_{Y_{(3)}}(y) = \frac{4!}{2!1!} \left(\frac{y}{5}\right)^2 \left(1 - \frac{y}{5}\right) \cdot \frac{1}{5} = \frac{12}{5} \frac{y^2}{25} \frac{5 - y}{5} = \frac{12y^2(5 - y)}{625}$$

and the corresponding cdf is

$$F_{Y_{(3)}}(y) = \int_0^y \frac{60t^2 - 12t^3}{625} dt = \frac{1}{625} [20y^3 - 3y^4] = \frac{20}{625} y^3 - \frac{3}{625} y^4$$

then

$$P(Y_{(3)} > 2) = 1 - \frac{20 \cdot 8 - 3 \cdot 16}{625} = \frac{513}{625} = 0.8208.$$

6. Five cards are dealt from a standard poker deck with 52 cards. Let  $X$  be the number of aces received (total of 4 aces in the deck) and  $Y$  the number of kings (total of 4 kings in the deck). Compute  $P(X = 2 | Y = 2)$ . Write a decimal number with three decimal places.

**Solution:**

$$P(X = 2 | Y = 2) = \frac{P(X = 2, Y = 2)}{P(Y = 2)} = \frac{\binom{4}{2} \binom{4}{2} \binom{44}{1}}{\binom{52}{5}} / \frac{\binom{4}{2} \binom{48}{3}}{\binom{52}{5}} = 0.015$$

7. A newly formed life insurance company has underwritten term policies with 120 women between the age of 40 and 44. Suppose that each woman has a 0.007 probability of dying during the next calendar year and that each death requires the company to pay out AUD 50,000 in benefits. What is the approximated probability (using the Poisson approximation) that the company will have to

pay at least AUD 150,000 in benefits next year? Write a decimal number with three decimal places.

**Solution:** Let  $X$  be the number of policy-holders who will die next year. Since  $n = 120$ ,  $p = 0.007$ , then  $\lambda = 0.84$ .

Then  $P(\text{company will pay AUD 150,000}) = P(X \geq 3) = 1 - P(X \leq 2) = 0.05334516$

8. Consider a standard normal random variable  $Z \sim N(0, 1)$ . Compute  $P(-0.44 < Z < 1.33)$ . Write a decimal number with three decimal places.

**Solutions:**

```
pnorm(1.33) - pnorm(-0.44)
#0.5782723
```

9. According to the Guardian Essential poll, the public support for the Indigenous voice to parliament has declined and in April 2023 59% of the Australian population with right to vote was in favour of it. In preparation for the referendum, a politician in support of the Voice to the parliament would like to organise a meeting with a large number of people against it, to try to change their intention to vote. What is the probability that the politician needs a meeting with at least 65 people to have 25 people who intend to vote no in the referendum (i.e. are currently against the Voice to parliament)? Consider people coming to the meeting as independent. Write a decimal number with three decimal places.

**Solution**

The probability of success is  $p = 1 - 0.59$  because the success in this case is to be for the no at the referendum.

Then the number of trials (people) needed to find 25 successes (people for the no) follow a negative binomial distribution  $NB(25, 0.41)$ .

And we can write  $P(X \geq 65) = 1 - P(X \leq 64)$ .

```
p <- 1-0.59
r <- 25

negbinom <- function(k,r,p){
  vec <- c()
  for(i in 1:length(k)){
    vec[i] <- choose(k[i]-1,r-1) * p^r *
```

```
(1-p)^(k[i]-r)
```

```
}
```

```
return(vec)
```

```
}
```

```
1-sum(negbinom(1:64,r,p))
```

```
# 0.3315857
```

10. The negative binomial distribution can be seen as a generalization of

**Solution:**

- the Poisson distribution
- the binomial distribution
- the geometric distribution (correct)
- the normal distribution